Zeta Function

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1 Definition

Definition 1.1. *The Riemann Zeta function is defined as*

$$\zeta(z) = \sum_{k=1}^{\infty} \frac{1}{k^z}$$

We have already shown that [1]

$$\zeta(2) = \sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} \tag{1}$$

2 Bernoulli Numbers

Examining the Taylor expansion of $\frac{x}{e^x-1}$, this will be required for the evaluation of the even integer values of ζ.

Definition 2.1. *The Bernoulli numbers B_n are defined as the coefficients of the Taylor expansion of the following function*

$$\boxed{\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n}$$

Theorem 2.1.

$$\sum_{k}^{n} \binom{n+1}{k} B_k = 0 \quad \forall n > 0, \ \textit{with } B_0 = 1 \tag{2}$$

Proof

Starting from definition 2.1

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

[1] See Gamma function document

Using the Taylor expansion of e^x

$$\frac{x}{\sum_{k=1}^{\infty} \frac{x^k}{k!}} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$x = \sum_{k=1}^{\infty} \frac{x^k}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$
$$1 = \sum_{k=1}^{\infty} \frac{x^{k-1}}{k!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x$$

Adjusting the indicies

$$1 = \sum_{k=0}^{\infty} \frac{x^k}{(k+1)!} \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Using Cauchy's product formula for infinite sums [2]

$$1 = \sum_{n=0}^{\infty} \underbrace{\sum_{k=0}^{n} \underbrace{\left(\frac{B_k}{k!} x^k\right)}_{a_k} \underbrace{\left(\frac{x^{n-k}}{(n-k+1)!}\right)}_{b_{n-k}}}_{c_n}$$

Simplifying

$$1 = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} \sum_{k=0}^{n} \binom{n+1}{k} B_k x^k x^{n-k}$$
$$1 = \sum_{n=0}^{\infty} \frac{x^n}{(n+1)!} \sum_{k=0}^{n} \binom{n+1}{k} B_k$$

Comparing coefficients of powers of x of both sides we get

$$n = 0, \quad 1 = B_0$$
$$n \neq 0, \quad 0 = \sum_{k=0}^{n} \binom{n+1}{k} B_k$$

2.1 First few Bernoulli numbers

Using equation 2 with $n = 1$ implies $\binom{2}{0} B_0 + \binom{2}{1} B_1 = 0 = 1 + 2B_1$ which implies

$$B_1 = -\frac{1}{2} \tag{3}$$

[2] $\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty} c_n$ where $c_n = \sum_{k=0}^{n} a_k b_{n-k}$. See Mertens' Theorem (proof in Bernoulli Numbers document).

Using equation 2 with $n = 2$ implies $\binom{3}{0}B_0 + \binom{3}{1}B_1 + \binom{3}{2}B_2 = 0 = 1 - \frac{3}{2} + 3B_2$ which implies

$$B_2 = \frac{1}{6} \tag{4}$$

Similarly when we apply this equation for increasing values of n

$$B_3 = 0, B_4 = -\frac{1}{30}, B_5 = 0, B_6 = \frac{1}{42}, \dots \tag{5}$$

Theorem 2.2. *In fact we can show that all odd Bernoulli numbers after n=1 are 0*

$$B_n = 0 \quad \forall \text{ odd } n > 1 \tag{6}$$

Proof

Starting from definition 2.1

$$\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n}{n!} x^n$$

Removing the first two terms of the sum

$$\frac{x}{e^x - 1} = B_0 + B_1 x + \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

Rearranging

$$\frac{x}{e^x - 1} - 1 + \frac{x}{2} = \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$
$$\frac{x + xe^x}{2e^x - 2} - 1 = \sum_{n=2}^{\infty} \frac{B_n}{n!} x^n$$

If we can show that the RHS is an even function then we have established the proof

$$y(x) = \frac{x + xe^x}{2e^x - 2} - 1$$
$$y(-x) = \frac{-x + -xe^{-x}}{2e^{-x} - 2} - 1$$

Simplfying

$$y(-x) = \frac{-x - xe^{-x}}{2e^{-x} - 2} \cdot \frac{e^x}{e^x} - 1$$
$$y(-x) = \frac{-xe^x - x}{2 - 2e^x} - 1$$
$$y(-x) = \frac{x + xe^x}{2e^x - 2} - 1$$
$$y(-x) = y(x)$$
$$B_n = 0 \quad \forall \text{ odd } n > 1$$

3 Even integer values $\zeta(2n)$

While the odd integer values of $\zeta(n)$ do not have a closed form expression, the even values do.

Theorem 3.1. *We can express all even integer values of ζ using Bernoulli numbers*

$$\zeta(2n) = (-1)^{n+1}\frac{(2\pi)^{2n}B_{2n}}{2\cdot(2n)!} \tag{7}$$

Proof

Starting from the series representation [3] *of cotangent*

$$\pi x cot(\pi x) = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Using the complex exponential representation [4] *of cotangent*

$$i\pi x + \frac{2i\pi x}{e^{2i\pi x}-1} = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Using the Bernoulli numbers definition 2.1

$$i\pi x + \sum_{n=0}^{\infty}\frac{B_n}{n!}(2i\pi x)^n = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Rearranging and using the first two values of the Bernoulli numbers

$$i\pi x + B_0 + B_1(2i\pi x) + \sum_{n=2}^{\infty}\frac{B_n}{n!}(2i\pi x)^n = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$
$$i\pi x + 1 + -i\pi x + \sum_{n=2}^{\infty}\frac{B_n}{n!}(2i\pi x)^n = 1 - 2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$
$$\sum_{n=2}^{\infty}\frac{B_n}{n!}(2i\pi)^n x^n = -2\sum_{k=1}^{\infty}\zeta(2k)x^{2k}$$

Using the fact the the odd Bernoulli numbers after B_1 are 0

$$\sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}(-1)^k(2\pi)^{2k}x^{2k} = \sum_{k=1}^{\infty}-2\zeta(2k)x^{2k}$$

Comparing coefficients

$$\zeta(2k) = \frac{B_{2k}}{2\cdot(2k)!}(-1)^{k+1}(2\pi)^{2k} \quad \forall k \in \mathbb{N}_{>0}$$

[3] See Euler's Sine Product Formula document

[4] $\cot(x) = i + \frac{2i}{e^{2ix}-1}$. See Basic trig functions document

3.1 The first few even integer values of ζ

$$\zeta(2) = \frac{\pi^2}{6}, \zeta(4) = \frac{\pi^4}{90}, \zeta(6) = \frac{\pi^6}{945}, \zeta(8) = \frac{\pi^8}{9450}, \zeta(10) = \frac{\pi^{10}}{93555},$$
$$\zeta(12) = \frac{691}{638512875}\pi^{12}, \zeta(14) = \frac{2}{18243225}\pi^{14}, \zeta(16) = \frac{3617}{325641566250}\pi^{16},$$
$$\zeta(18) = \frac{43867}{38979295480125}\pi^{18}, \zeta(20) = \frac{174611}{1531329465290625}\pi^{20}, \zeta(22) = \frac{155366}{13447856940643125}\pi^{22},$$
$$\zeta(24) = \frac{236364091}{20191957196375652187 5}\pi^{24}, \ldots$$

4 Related functions

4.1 Dirichlet Eta Function

Definition 4.1. *The Dirichlet Eta function is defined as*

$$\eta(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^z} = 1 - \frac{1}{2^z} + \frac{1}{3^z} - \frac{1}{4^z} + ...$$

Theorem 4.1.

$$\boxed{\eta(z) = (1 - 2^{1-s})\zeta(z)} \tag{8}$$

Proof

Starting from the definition of $\eta(z)$

$$\eta(z) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^z} = \sum_{k=1}^{\infty} \frac{1}{k^z} - 2\sum_{k=1}^{\infty} \frac{1}{(2k)^z}$$

Using the definition of $\zeta(z)$

$$\eta(z) = \zeta(z) - \frac{2}{2^z}\zeta(z)$$

$$\eta(z) = (1 - 2^{1-z})\zeta(z)$$

4.1.1 Values

$$\eta(1) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = ln(2) \tag{9}$$ [5]

$$\eta(2) = (1 - 2^{1-2}) \cdot \zeta(2) = \frac{1}{2} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{12} \tag{10}$$

[5] This can be seen by taking the Taylor-Expansion of $\ln(x+1)$ and evaluating at $x = 1$

4.2 Dirichlet Lambda Function

Definition 4.2. *The Dirichlet Lambda function is defined as*

$$\lambda(z) = \sum_{k=0}^{\infty} \frac{1}{(2k+1))^z} = 1 + \frac{1}{3^z} + \frac{1}{5^z} + \frac{1}{7^z} + ...$$

Theorem 4.2.

$$\boxed{\lambda(z) = (1 - 2^{-s})\zeta(z)} \tag{11}$$

Proof

Starting from the definition of $\lambda(z)$

$$\lambda(z) = \sum_{k=0}^{\infty} \frac{1}{(2k+1))^z} = \sum_{k=1}^{\infty} \frac{1}{k^z} - \sum_{k=1}^{\infty} \frac{1}{(2k)^z}$$

Using the definition of $\zeta(z)$

$$\lambda(z) = \zeta(z) - \frac{1}{2^z}\zeta(z)$$

$$\lambda(z) = (1 - 2^{-z})\zeta(z)$$

4.2.1 Values

$$\lambda(2) = (1 - 2^{-2}) \cdot \zeta(2) = \frac{3}{4} \cdot \frac{\pi^2}{6} = \frac{\pi^2}{8} \tag{12}$$

5 Integral Forms

5.1 Gamma with Zeta functions

Theorem 5.1. *See the Gamma function document for the definition and properties of the Gamma function.*

$$\boxed{\Gamma(z)\zeta(z) = \int_0^\infty \frac{u^{z-1}}{e^u - 1} du} \tag{13}$$

Proof

Using $\Gamma(z) = n^z \int_0^\infty e^{-nu} u^{z-1} du$ *from the Gamma Function document and rearranging*

$$\Gamma(z)\frac{1}{n^z} = \int_0^\infty e^{-nu} u^{z-1} du$$

Let n *be all integers and take the sum of all equalities*

$$\sum_{n=1}^\infty \Gamma(z)\frac{1}{n^z} = \Gamma(z)\zeta(z) = \sum_{n=1}^\infty \int_0^\infty e^{-nu} u^{z-1} du$$

Bringing the summation into the integral since this integral converges uniformly

$$\Gamma(z)\zeta(z) = \int_0^\infty \Big(\sum_{n=1}^\infty e^{-nu}\Big) u^{z-1} du$$

Notice that $\sum_{n=1}^\infty e^{-nu}$ *is a geometric series if* $u > 0$

$$\Gamma(z)\zeta(z) = \int_0^\infty \Big(\frac{e^{-u}}{1 - e^{-u}}\Big) u^{z-1} du$$

Simplfiying by multiplying top and bottom by e^u

$$\Gamma(z)\zeta(z) = \int_0^\infty \frac{u^{z-1}}{e^u - 1} du$$

6 Jacobi Theta Function

6.1 Definition

Definition 6.1. *The Jacobi Theta function is defined as*

$$\Theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x}$$

Theorem 6.1. *The Jacobi Theta function has the following functional form*

$$\Theta(x) = \frac{1}{\sqrt{x}}\Theta(\frac{1}{x}) \tag{14}$$

Proof

Using the Poisson Summation Formula [6]

$$\Theta(x) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 x} = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi y^2 x} e^{-2\pi i k y} dy = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi y^2 x - 2\pi i k y} dy$$

Re-writting the exponent

$$-\pi y^2 x - 2\pi i k y = -\pi x(y^2 - 2\pi i \frac{k}{x} y) = -\pi x(y + i\frac{k}{x})^2 - \pi \frac{k^2}{x}$$

Plugging in the factored exponent, we get

$$\Theta(x) = \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi x(y+i\frac{k}{x})^2 - \pi \frac{k^2}{x}} dy = \sum_{k=-\infty}^{\infty} e^{-\pi \frac{k^2}{x}} \int_{-\infty}^{\infty} e^{-\pi x(y+i\frac{k}{x})^2} dy$$

After a change of variables $z = y + i\frac{k}{x}$ we can use the Guassian integral identity as proved before, refer to the Guassian integral document, $\int_{-\infty}^{\infty} e^{-az^2} dz = \sqrt{\frac{\pi}{a}}$, we get:

$$\Theta(x) = \sum_{k=-\infty}^{\infty} e^{-\pi \frac{k^2}{x}} \sqrt{\frac{\pi}{x\pi}} = \frac{1}{\sqrt{x}}\Theta(\frac{1}{x})$$

7 Functional Equations

7.1 First Symmetric Form

Theorem 7.1 (Riemann's Symmetric Functional Equation)**.** *The completed zeta function $\xi(s)$ satisfies the symmetric functional equation:*

$$\boxed{\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \pi^{-\frac{1-s}{2}}\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s)} \tag{15}$$

[6] $\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \hat{f}(n)$ is the Poisson summation formula with Fourier transform $\hat{f}(k) = \int_{-\infty}^{\infty} f(x)e^{-2\pi i k x} dx$. See Fourier Series and Integrals notes.

Proof. Recall the integral definition of the Gamma function:

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty t^{\frac{s}{2}-1} e^{-t}\, dt$$

Making the substitution $t = \pi n^2 x$ (so $dt = \pi n^2 dx$) for $n \geq 1$:

$$\Gamma\left(\frac{s}{2}\right) = \int_0^\infty (\pi n^2 x)^{\frac{s}{2}-1} e^{-\pi n^2 x} \pi n^2\, dx = \pi^{\frac{s}{2}} n^s \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x}\, dx$$

Rearranging yields:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) n^{-s} = \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x}\, dx$$

Summing both sides over all positive integers $n = 1, 2, 3, \ldots$ for $\mathrm{Re}(s) > 1$:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \sum_{n=1}^\infty n^{-s} = \sum_{n=1}^\infty \int_0^\infty x^{\frac{s}{2}-1} e^{-\pi n^2 x}\, dx$$

By Fubini-Tonelli theorem, we can interchange the sum and integral:

$$\pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \int_0^\infty x^{\frac{s}{2}-1} \omega(x)\, dx$$

where $\omega(x) := \sum_{n=1}^\infty e^{-\pi n^2 x} = \frac{\Theta(x)-1}{2}$.

From Jacobi's theta transformation $\Theta(x) = \frac{1}{\sqrt{x}}\Theta\left(\frac{1}{x}\right)$, the function $\omega(x)$ satisfies:

$$\omega(x) = \frac{1}{\sqrt{x}}\omega\left(\frac{1}{x}\right) + \frac{1}{2\sqrt{x}} - \frac{1}{2}$$

Now split the integral into two regions $[0, 1]$ and $[1, \infty)$:

$$\int_0^\infty x^{\frac{s}{2}-1} \omega(x)\, dx = \int_0^1 x^{\frac{s}{2}-1} \omega(x)\, dx + \int_1^\infty x^{\frac{s}{2}-1} \omega(x)\, dx$$

Substitute the modular transformation of $\omega(x)$ into the integral over $[0, 1]$:

$$\begin{aligned}
\int_0^1 x^{\frac{s}{2}-1} \omega(x)\, dx &= \int_0^1 x^{\frac{s}{2}-1} \left(\frac{1}{\sqrt{x}}\omega\left(\frac{1}{x}\right) + \frac{1}{2x^{1/2}} - \frac{1}{2}\right) dx \\
&= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{2}\left[\frac{x^{s/2-1/2}}{s/2-1/2}\right]_0^1 - \frac{1}{2}\left[\frac{x^{s/2}}{s/2}\right]_0^1 \\
&= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{s-1} - \frac{1}{s} \\
&= \int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx + \frac{1}{s(s-1)}
\end{aligned}$$

In the integral $\int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right)\, dx$, substitute $u = 1/x$ ($dx = -u^{-2} du$):

$$\int_0^1 x^{\frac{s-1}{2}-1} \omega\left(\frac{1}{x}\right) dx = \int_\infty^1 u^{-\left(\frac{s-1}{2}-1\right)} \omega(u) \left(-\frac{du}{u^2}\right) = \int_1^\infty u^{\frac{1-s}{2}-1} \omega(u)\, du$$

Combining both parts, we obtain the fundamental representation:

$$\xi(s) := \pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \left(x^{\frac{s}{2}-1} + x^{\frac{1-s}{2}-1}\right)\omega(x)\,dx \tag{16}$$

Because $\omega(x) \sim e^{-\pi x}$ as $x \to \infty$, the integral converges absolutely and uniformly for all $s \in \mathbb{C}$, providing the analytic continuation of $\zeta(s)$ to a meromorphic function on $\mathbb{C}$ with a single simple pole at $s = 1$.

Moreover, replacing s with $1 - s$ leaves the right-hand side completely invariant:

$$\xi(1-s) = \frac{1}{(1-s)(-s)} + \int_1^\infty \left(x^{\frac{1-s}{2}-1} + x^{\frac{s}{2}-1}\right)\omega(x)\,dx = \xi(s)$$

which proves the theorem. □

7.2 Second Asymmetric Form

Theorem 7.2 (Reflection Form of the Functional Equation)**.** *For all* $s \in \mathbb{C} \setminus \{0, 1\}$*:*

$$\boxed{\zeta(s) = 2^s\pi^{s-1}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)\zeta(1-s)} \tag{17}$$

Proof. From the first functional equation:

$$\zeta(s) = \pi^{s-\frac{1}{2}}\frac{\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)}\zeta(1-s)$$

We simplify the ratio $\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}$ using two classical identities of the Gamma function:

1. **Legendre's Duplication Formula**:

$$\Gamma\left(\frac{s}{2}\right)\Gamma\left(\frac{s+1}{2}\right) = 2^{1-s}\sqrt{\pi}\,\Gamma(s) \implies \frac{1}{\Gamma(s/2)} = \frac{2^{s-1}}{\sqrt{\pi}}\frac{\Gamma\left(\frac{s+1}{2}\right)}{\Gamma(s)}$$

2. **Euler's Reflection Formula**:

$$\Gamma\left(\frac{1-s}{2}\right)\Gamma\left(\frac{1+s}{2}\right) = \frac{\pi}{\sin\left(\pi\frac{1-s}{2}\right)} = \frac{\pi}{\cos\left(\frac{\pi s}{2}\right)}$$

and for $\Gamma(s)$:

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)} \implies \frac{1}{\Gamma(s)} = \frac{\sin(\pi s)}{\pi}\Gamma(1-s) = \frac{2\sin\left(\frac{\pi s}{2}\right)\cos\left(\frac{\pi s}{2}\right)}{\pi}\Gamma(1-s)$$

Combining these gives:

$$\begin{aligned}
\frac{\Gamma\left(\frac{1-s}{2}\right)}{\Gamma\left(\frac{s}{2}\right)} &= \Gamma\left(\frac{1-s}{2}\right)\cdot\frac{2^{s-1}}{\sqrt{\pi}}\frac{\Gamma\left(\frac{s+1}{2}\right)}{\Gamma(s)} \\
&= \frac{2^{s-1}}{\sqrt{\pi}}\left[\Gamma\left(\frac{1-s}{2}\right)\Gamma\left(\frac{1+s}{2}\right)\right]\cdot\frac{1}{\Gamma(s)} \\
&= \frac{2^{s-1}}{\sqrt{\pi}}\left[\frac{\pi}{\cos\left(\frac{\pi s}{2}\right)}\right]\cdot\left[\frac{2\sin\left(\frac{\pi s}{2}\right)\cos\left(\frac{\pi s}{2}\right)}{\pi}\Gamma(1-s)\right] \\
&= 2^s\pi^{-\frac{1}{2}}\sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)
\end{aligned}$$

Multiplying by $\pi^{s-1/2}\zeta(1-s)$ yields:

$$\zeta(s) = \pi^{s-\frac{1}{2}} \cdot 2^s \pi^{-\frac{1}{2}} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s)$$

□

8 Zeros of the Riemann Zeta Function and the Prime Numbers

8.1 The Euler Product: The Bridge to Prime Numbers

The profound connection between the Riemann zeta function and prime numbers was discovered by Leonhard Euler in 1737. It shows that the analytical behavior of $\zeta(s)$ directly encodes the distribution of prime numbers.

Theorem 8.1 (Euler Product Formula)**.** *For any complex number s with $\mathrm{Re}(s) > 1$, the zeta function can be expressed as an absolutely convergent infinite product over all prime numbers:*

$$\boxed{\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}} \tag{18}$$

Proof. Consider the series definition $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$ for $\mathrm{Re}(s) > 1$. For each prime p, the term p^{-s} satisfies $|p^{-s}| = p^{-\mathrm{Re}(s)} < 1$. Expanding each geometric factor:

$$\frac{1}{1-p^{-s}} = 1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \cdots = \sum_{k=0}^{\infty} \frac{1}{p^{ks}}$$

Now consider the finite product over the first N primes $p_1, p_2, \ldots, p_N$:

$$\begin{aligned}\prod_{j=1}^{N} \frac{1}{1-p_j^{-s}} &= \left(\sum_{k_1=0}^{\infty} \frac{1}{p_1^{k_1 s}}\right)\left(\sum_{k_2=0}^{\infty} \frac{1}{p_2^{k_2 s}}\right) \cdots \left(\sum_{k_N=0}^{\infty} \frac{1}{p_N^{k_N s}}\right) \\ &= \sum_{k_1=0}^{\infty} \sum_{k_2=0}^{\infty} \cdots \sum_{k_N=0}^{\infty} \frac{1}{(p_1^{k_1} p_2^{k_2} \cdots p_N^{k_N})^s}\end{aligned}$$

By the **Fundamental Theorem of Arithmetic**, every positive integer $n \geq 1$ has a unique factorization as a product of prime powers $n = p_1^{k_1} p_2^{k_2} \cdots$. Therefore, multiplying out the terms produces the sum $\sum \frac{1}{n^s}$ over all positive integers n whose prime factors are all among $\{p_1, \ldots, p_N\}$:

$$\zeta(s) - \prod_{j=1}^{N} \frac{1}{1-p_j^{-s}} = \sum_{\substack{n > p_N \\ \exists q > p_N \text{ s.t. } q \mid n}} \frac{1}{n^s}$$

Because $\sum_{n=1}^{\infty} |n^{-s}| = \sum_{n=1}^{\infty} n^{-\mathrm{Re}(s)} < \infty$, the tail of the series vanishes as $N \to \infty$:

$$\left|\zeta(s) - \prod_{j=1}^{N} \frac{1}{1-p_j^{-s}}\right| \leq \sum_{n=p_N+1}^{\infty} \frac{1}{n^{\mathrm{Re}(s)}} \xrightarrow[N\to\infty]{} 0$$

which completes the proof. □

Corollary 8.1.1 (No Zeros in the Half-Plane $\mathrm{Re}(s) > 1$)**.** *The Riemann zeta function has no zeros in the region* $\mathrm{Re}(s) > 1$.

Proof. Taking the complex logarithm of the Euler product for $\mathrm{Re}(s) > 1$:

$$\ln \zeta(s) = -\sum_p \ln(1 - p^{-s}) = \sum_p \sum_{m=1}^{\infty} \frac{1}{m\, p^{ms}}$$

Since $\sum_p \sum_{m=1}^{\infty} \frac{1}{mp^{m\,\mathrm{Re}(s)}} \leq \sum_p \frac{2}{p^{\mathrm{Re}(s)}} < \infty$, this series converges absolutely. Consequently, $\ln \zeta(s)$ is finite everywhere on $\mathrm{Re}(s) > 1$. Therefore:

$$|\zeta(s)| = \exp(\mathrm{Re}(\ln \zeta(s))) > 0$$

meaning $\zeta(s) \neq 0$ for all $\mathrm{Re}(s) > 1$. □

8.2 Trivial Zeros and the Critical Strip

With the half-plane $\mathrm{Re}(s) > 1$ established as zero-free, we examine the remaining regions using the reflection functional equation (Theorem 7.2):

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1 - s)\zeta(1 - s) \tag{19}$$

8.2.1 Trivial Zeros

Consider negative even integers $s = -2m$ for $m \in \{1, 2, 3, \dots\}$:

(i) The trigonometric factor vanishes: $\sin\left(\frac{\pi(-2m)}{2}\right) = \sin(-\pi m) = 0$.

(ii) The gamma factor $\Gamma(1 - (-2m)) = \Gamma(1 + 2m) = (2m)!$ is a positive real number, hence finite and non-zero.

(iii) The zeta factor $\zeta(1 - (-2m)) = \zeta(2m + 1)$ is evaluated at an odd integer $2m + 1 \geq 3 > 1$, where the Euler product guarantees $\zeta(2m + 1) > 1 \neq 0$.

Therefore, every negative even integer is a zero of $\zeta(s)$:

$$\boxed{\zeta(-2m) = 0 \quad \text{for all } m \in \{1, 2, 3, \dots\}} \tag{20}$$

These are the **trivial zeros** of the Riemann zeta function.

Remark 8.2 (Values at Negative Odd Integers)**.** *At negative odd integers* $s = -(2m-1)$*, the sine factor does not vanish:* $\sin\left(-\frac{(2m-1)\pi}{2}\right) = (-1)^m \neq 0$*. Using the functional equation together with Euler's formula for* $\zeta(2m)$ *(Theorem* **??***), we obtain:*

$$\zeta(-(2m - 1)) = -\frac{B_{2m}}{2m} \neq 0 \tag{21}$$

In particular:

$$\zeta(0) = -\frac{1}{2}, \quad \zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = \frac{1}{120}, \quad \zeta(-5) = -\frac{1}{252}$$

8.2.2 The Critical Strip

Combining our results:

1. **Region** $\mathrm{Re}(s) > 1$: $\zeta(s) \neq 0$ by the Euler product (Corollary 8.1.1).

2. **Region** $\mathrm{Re}(s) < 0$: By the functional equation (19), the only zeros for $\mathrm{Re}(s) < 0$ are the trivial zeros at $s = -2, -4, -6, \ldots$, since $\zeta(1-s) \neq 0$ for $\mathrm{Re}(1-s) > 1$.

3. **The Critical Strip**: All non-trivial zeros must lie strictly within the strip:

$$\mathcal{S} = \{s \in \mathbb{C} : 0 \leq \mathrm{Re}(s) \leq 1\} \tag{22}$$

4. **Zero-Free Boundary**: In 1896, Jacques Hadamard and Charles Jean de la Vallée Poussin independently proved that $\zeta(s)$ has no zeros on the line $\mathrm{Re}(s) = 1$:

$$\zeta(1+it) \neq 0 \quad \text{for all } t \in \mathbb{R}$$

   By the reflection symmetry $s \leftrightarrow 1-s$, $\zeta(0+it) \neq 0$ as well. This zero-free boundary is mathematically equivalent to the **Prime Number Theorem** ($\pi(x) \sim \frac{x}{\ln x}$).

Consequently, all non-trivial zeros lie strictly in the interior $0 < \mathrm{Re}(s) < 1$.

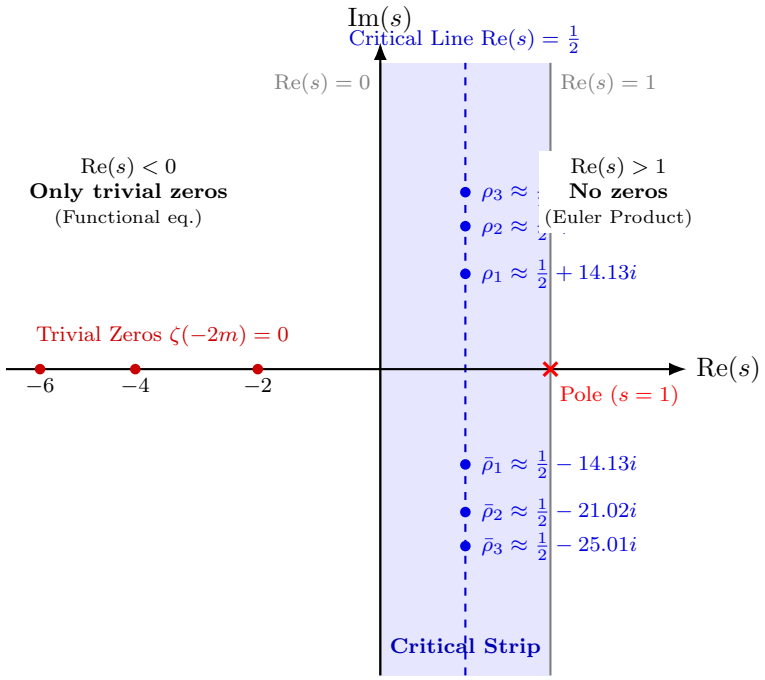


Figure 1: The complex s-plane: trivial zeros on the negative real axis, the pole at $s = 1$, the critical strip $0 < \mathrm{Re}(s) < 1$, and non-trivial zeros lying symmetrically on the critical line $\mathrm{Re}(s) = \frac{1}{2}$.

8.3 The Riemann Xi Function and Critical Line Symmetry

To study the non-trivial zeros on the critical line, Riemann introduced the completed function $\xi(s)$ in his 1859 paper:

$$\xi(s) := \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{23}$$

The prefactor $\frac{1}{2}s(s-1)$ cancels the simple pole of $\zeta(s)$ at $s=1$ and the pole of $\Gamma(s/2)$ at $s=0$. Consequently, $\xi(s)$ is an **entire function** of order 1.

From Theorem 7.1, $\xi(s)$ satisfies the perfect reflection symmetry:

$$\xi(s) = \xi(1-s) \tag{24}$$

Notice that the zeros of $\xi(s)$ are precisely the non-trivial zeros of $\zeta(s)$ (the trivial zeros are cancelled by the poles of $\Gamma(s/2)$).

Theorem 8.3 (The Real-Valued Function $\Xi(t)$)**.** *Parameterizing the critical line by $s = \frac{1}{2} + it$ for $t \in \mathbb{R}$, define:*

$$\Xi(t) := \xi\left(\frac{1}{2} + it\right) \tag{25}$$

*Then $\Xi(t)$ is an **even**, **purely real-valued** function for all real t:*

$$\Xi(-t) = \Xi(t) \quad \textit{and} \quad \Xi(t) \in \mathbb{R} \quad (\forall t \in \mathbb{R})$$

Proof. Using the functional equation $\xi(s) = \xi(1-s)$ with $s = \frac{1}{2} + it$:

$$\Xi(-t) = \xi\left(\frac{1}{2} - it\right) = \xi\left(1 - \left(\frac{1}{2} + it\right)\right) = \xi\left(\frac{1}{2} + it\right) = \Xi(t)$$

which proves that $\Xi(t)$ is an even function.

To prove that $\Xi(t)$ is real-valued, observe that for real s, $\zeta(s)$, $\Gamma(s/2)$, and $s(s-1)$ are all real. By the Schwarz reflection principle, $\overline{\xi(s)} = \xi(\bar{s})$ for all $s \in \mathbb{C}$. Evaluating at $s = \frac{1}{2} + it$:

$$\overline{\Xi(t)} = \overline{\xi\left(\frac{1}{2} + it\right)} = \xi\left(\overline{\frac{1}{2} + it}\right) = \xi\left(\frac{1}{2} - it\right) = \Xi(-t) = \Xi(t)$$

Since $\overline{\Xi(t)} = \Xi(t)$, $\Xi(t)$ must be real for all $t \in \mathbb{R}$. □

Remark 8.4 (Significance for Zeros)**.** *Theorem 8.3 converts the two-dimensional problem of locating complex zeros of $\zeta(s)$ on the critical line into the one-dimensional problem of finding the **real roots of a real-valued curve** $\Xi(t)$.*

In 1914, G. H. Hardy used this real-variable formulation to prove that $\Xi(t)$ changes sign infinitely many times, establishing:

> ***Hardy's Theorem (1914):** There are infinitely many non-trivial zeros of $\zeta(s)$ lying exactly on the critical line $\mathrm{Re}(s) = \frac{1}{2}$.*

In 1989, J. B. Conrey proved that at least 40.77% of all non-trivial zeros lie strictly on the critical line.

8.4 The Riemann Hypothesis

The Riemann Hypothesis (Bernhard Riemann, 1859)

All non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the **critical line**:

$$\boxed{\mathrm{Re}(s) = \frac{1}{2}} \tag{26}$$

Equivalently, all roots of the Riemann Ξ-function $\Xi(t)$ are purely real:

$$\Xi(t) = 0 \implies t \in \mathbb{R} \tag{27}$$

The first five non-trivial zeros on the critical line have imaginary parts:

$$\begin{aligned}
\rho_1 &= \frac{1}{2} + i \cdot 14.13472514\ldots \\
\rho_2 &= \frac{1}{2} + i \cdot 21.02203964\ldots \\
\rho_3 &= \frac{1}{2} + i \cdot 25.01085758\ldots \\
\rho_4 &= \frac{1}{2} + i \cdot 30.42487613\ldots \\
\rho_5 &= \frac{1}{2} + i \cdot 32.93506159\ldots
\end{aligned}$$

To date, more than 10^{13} non-trivial zeros have been checked by computer algorithms, and every single one lies precisely on the critical line $\mathrm{Re}(s) = 1/2$. A general mathematical proof remains the most celebrated open problem in mathematics and is designated as a Clay Millennium Prize Problem.

8.5 The "Music of the Primes": Riemann's Explicit Formula

Riemann's ultimate goal in studying $\zeta(s)$ was to understand the distribution of prime numbers with exact precision.

8.5.1 Von Mangoldt and Chebyshev Functions

To link primes with analytic continuation, define the **von Mangoldt function**:

$$\Lambda(n) = \begin{cases} \ln p & \text{if } n = p^k \text{ for some prime } p \text{ and integer } k \geq 1 \\ 0 & \text{otherwise} \end{cases} \tag{28}$$

Taking the logarithmic derivative of the Euler product (Theorem 8.1):

$$\ln \zeta(s) = -\sum_{p} \ln(1 - p^{-s})$$

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{p} \frac{p^{-s} \ln p}{1 - p^{-s}} = \sum_{p} \sum_{k=1}^{\infty} (\ln p)\, p^{-ks} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}$$

The cumulative sum $\psi(x) = \sum_{n \le x} \Lambda(n)$ is **Chebyshev's second function**. Counting primes via $\psi(x)$ is weighted by $\ln p$, giving $\psi(x) \approx \sum_{p \le x} \ln p \approx \pi(x) \ln x$.

8.5.2 Riemann's Explicit Formula

By applying Perron's contour inversion formula to $-\frac{\zeta'(s)}{\zeta(s)}$ and evaluating the residues at the simple pole $s = 1$, the trivial zeros, and the non-trivial zeros ρ, Riemann derived his celebrated **Explicit Formula**:

Theorem 8.5 (Riemann's Explicit Formula)**.** *For any non-integer $x > 1$:*

$$\boxed{\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \ln(2\pi) - \frac{1}{2}\ln(1 - x^{-2})} \tag{29}$$

where the sum ranges over all non-trivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$, paired as $\lim_{T\to\infty} \sum_{|\gamma|<T} \frac{x^\rho}{\rho}$.

8.5.3 The Harmonic Interpretation

Writing each non-trivial zero as $\rho = \beta + i\gamma$:

$$x^\rho = x^{\beta+i\gamma} = x^\beta e^{i\gamma \ln x} = x^\beta \left[\cos(\gamma \ln x) + i \sin(\gamma \ln x)\right] \tag{30}$$

Pairing conjugate zeros $\rho = \beta + i\gamma$ and $\bar{\rho} = \beta - i\gamma$ produces real oscillating harmonics:

$$\frac{x^\rho}{\rho} + \frac{x^{\bar{\rho}}}{\bar{\rho}} = 2\,\mathrm{Re}\left(\frac{x^{\beta+i\gamma}}{\beta + i\gamma}\right) = \frac{2x^\beta}{|\rho|}\cos\left(\gamma \ln x - \arg(\rho)\right)$$

Equation (29) decomposes the prime distribution into:

1. **The Baseline Trend** x: Produced by the residue at the pole $s = 1$. This corresponds to the smooth asymptotic distribution of primes $\psi(x) \sim x$ (the Prime Number Theorem).
2. **The Harmonic Oscillations** $-\sum_\rho \frac{x^\rho}{\rho}$: Each non-trivial zero ρ contributes a pure wave with frequency $\frac{\gamma}{2\pi}$ on a logarithmic scale $\ln x$. The primes are literally the **constructive interference** of these infinite waves!
3. **The Smooth Tail** $-\ln(2\pi) - \frac{1}{2}\ln(1 - x^{-2})$: Produced by the residues at the trivial zeros $s = -2, -4, \ldots$ and $s = 0$.

8.5.4 The Optimal Error Bound

The real part $\beta = \mathrm{Re}(\rho)$ controls the **amplitude** of the prime fluctuations (x^β).

- If any zero had $\beta > 1/2$, the fluctuations would blow up with amplitude x^β, creating massive, irregular swings in prime density.
- Under the **Riemann Hypothesis**, all zeros have $\beta = 1/2$, making every oscillation have amplitude exactly $\sqrt{x}$.

This establishes the ultimate equivalence:

Theorem 8.6 (Equivalence with the Prime Error Bound)**.** *The Riemann Hypothesis is mathematically equivalent to the sharpest possible error bound for prime counting:*

$$\boxed{\pi(x) = \mathrm{Li}(x) + O\left(\sqrt{x}\ln x\right)} \tag{31}$$

where $\mathrm{Li}(x) = \int_2^x \frac{dt}{\ln t}$ *is the logarithmic integral.*

The error $O(\sqrt{x}\ln x)$ is identical to the statistical fluctuation of an unbiased random walk of size x. Thus, the Riemann Hypothesis asserts that prime numbers are distributed with the maximum possible regularity permitted by arithmetic.